

# Econometrics

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Week 2 Notes

## Contents

|                                                   |          |
|---------------------------------------------------|----------|
| <b>1 Qualitative Variables as Predictors</b>      | <b>1</b> |
| 1.1 Importing the data . . . . .                  | 1        |
| 1.2 Indicator variables . . . . .                 | 3        |
| 1.3 Interpretation of parameters . . . . .        | 4        |
| <b>2 Variable Selection Procedures</b>            | <b>4</b> |
| 2.1 Searching over all possible subsets . . . . . | 5        |
| 2.2 Stepwise procedures . . . . .                 | 5        |
| 2.2.1 Forward Selection . . . . .                 | 5        |
| 2.2.2 Backward Selection . . . . .                | 6        |
| 2.2.3 Stepwise Selection . . . . .                | 7        |
| 2.3 Model Selection . . . . .                     | 7        |

## 1 Qualitative Variables as Predictors

In a linear regression model  $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$  what if some predictor  $X_j$  is **categorical**

Example is  $X_1 = \text{blood type(O,A,B,AB)}$

It makes no sense to write a model like:

$$Y = \beta_0 + \beta_1(\text{bloodtype}) + \epsilon$$

since blood type is not a number.

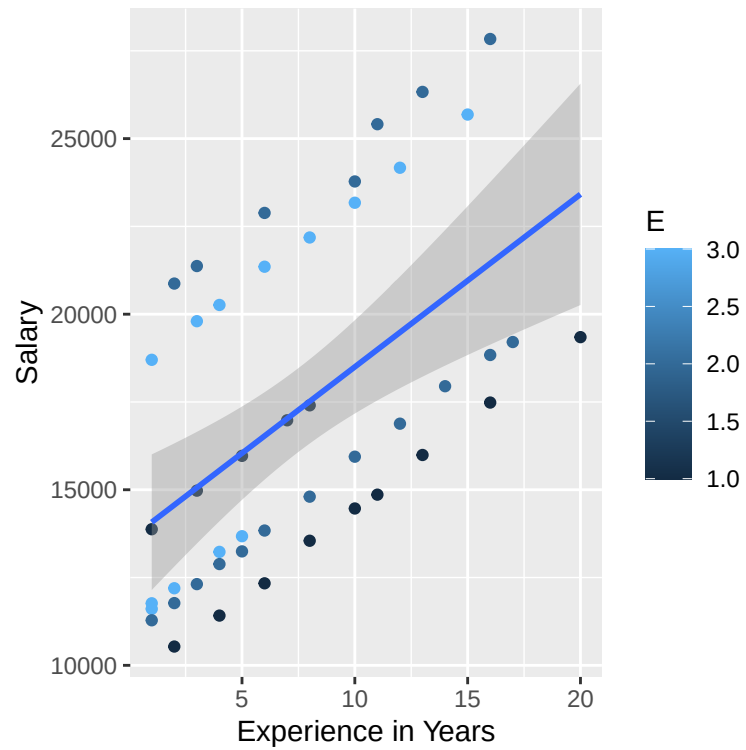
However many demographics (Gender, Marital Status) are categorical and can provide useful info for predicting the response variable  $Y$

How to represent the categorical variable numerically in a model the solution is to create an **indicator** or **dummy variable** for each category of the categorical variable.

### 1.1 Importing the data

R regards  $E = 1, 2, 3$  as numerical rather than categorical

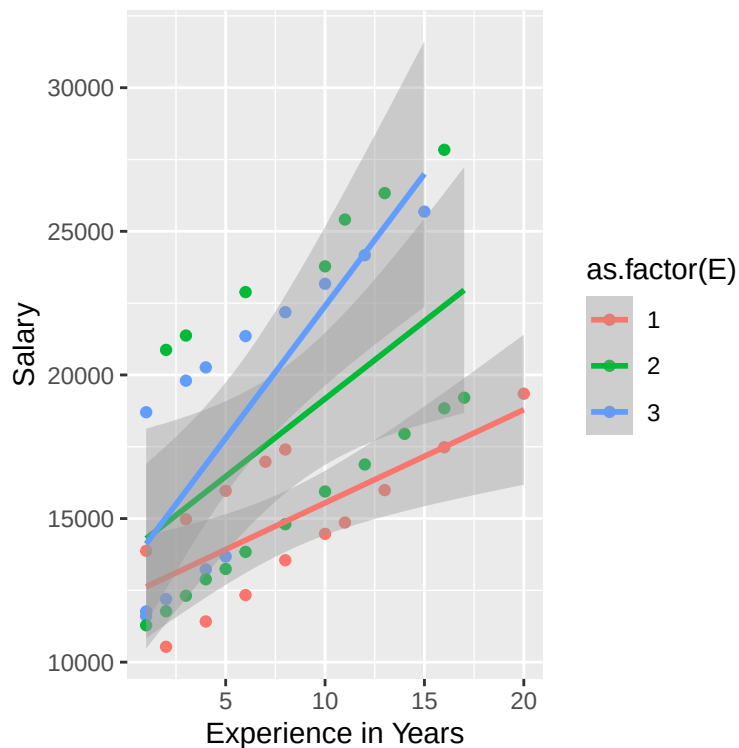
```
data <- read.csv("data.csv")
ggplot(data, aes(x = X, y = S, color = E)) +
  geom_point() + geom_smooth(method = "lm") +
  labs(x = "Experience in Years", y = "Salary")
```



The command `as.factor()` let R know that E is a categorical variable

```
data <- read.csv("data.csv")
ggplot(data, aes(x = X, y = S, color = as.factor(E))) +
  geom_point() + geom_smooth(method = "lm") +
  labs(x = "Experience in Years", y = "Salary")
```

## 'geom\_smooth()' using formula = 'y ~ x'



## 1.2 Indicator variables

They are also known as dummy variables.

The response variable  $S$  - Salary

Experience  $X$  is numerical

Education  $E$  is categorical

The three categories need 3 indicator variables:

$$E_{i1} = 1 \text{ if } i^{th} \text{ has high school diploma, } 0 \text{ otherwise}$$

$$E_{i2} = 1 \text{ if } i^{th} \text{ has B.A or B.S, } 0 \text{ otherwise}$$

$$E_{i3} = 1 \text{ if } i^{th} \text{ has advanced degree, } 0 \text{ otherwise}$$

One can fit the model:

$$S = \beta_0 + \beta_1 X + \delta_2 E_2 + \delta_3 E_3 + \epsilon$$

**A categorical predictor with  $c$  categories needs only  $c-1$  indicator variables.**

The mean response  $E[S]$  for the three education levels are:

| Education        | Indicator          | S                                |
|------------------|--------------------|----------------------------------|
| HS diploma       | $E_1 = E_2 = 0$    | $\beta_0 + \beta_1 X$            |
| Bachelors degree | $E_2 = 1, E_3 = 0$ | $\beta_0 + \delta_2 + \beta_1 X$ |
| Advanced degree  | $E_2 = 0, E_3 = 1$ | $\beta_0 + \delta_3 + \beta_1 X$ |

### 1.3 Interpretation of parameters

```
data <- read.csv("data.csv")
model <- lm(S ~ X + as.factor(E), data = data)
pander(summary(model))
```

|               | Estimate | Std. Error | t value | Pr(> t )  |
|---------------|----------|------------|---------|-----------|
| (Intercept)   | 10474    | 1305       | 8.024   | 5.186e-10 |
| X             | 548.6    | 107.6      | 5.1     | 7.695e-06 |
| as.factor(E)2 | 3221     | 1276       | 2.525   | 0.01544   |
| as.factor(E)3 | 4780     | 1423       | 3.36    | 0.001669  |

Table 3: Fitting linear model:  $S \sim X + \text{as.factor}(E)$

| Observations | Residual Std. Error | $R^2$  | Adjusted $R^2$ |
|--------------|---------------------|--------|----------------|
| 46           | 3622                | 0.4498 | 0.4104         |

From the model the predicted salary is:

$$\hat{S} = 10474 + 548X + 3221E_2 + 4780E_3$$

The model implies that on average:

- Each extra year of experience increases salary by approximately 548.
- Completing college increases salary by 3221
- Completing advanced degree increases salary by 4780.

## 2 Variable Selection Procedures

Some benefits of using variable selection include:

- Improved Model Performance
- Reduced Model Complexity
- Improved Interpretability
- Increased Generalization

Variable selection will allow us to produce a **parsimonious** model that is a simple model and contains only the necessary variables to achieve a good fit to the data.

## 2.1 Searching over all possible subsets

The most direct and ideal approach is to examine **all possible subsets** of potential outcomes.

What are all possible subsets. If there are 3 potential predictors  $X_1, X_2$  and  $X_3$  the models we could consider include:

- 1 predictor
- 2 predictors
- 3 predictors

## 2.2 Stepwise procedures

As its not practical to search all possible subsets when  $q$  is large here are a few commonly used algorithms aimed to find the best model without look at all possible subsets.

- Forward selection
- Backward elimination
- Stepwise selection

### 2.2.1 Forward Selection

The **Forward Selection** algorithm consider all candidate subsets consisting of one additional term beyond the current subset.

- It begins with an intercept only model
- At each iteration add predictor with the smallest p-value and then fit the new model
- Stop if: The new selected variables is not significant or all variables have been selected

The FS algorithm considers at most

$$q + q(q - 1) + \dots + 2 + 1 = \frac{q(q + 1)}{2}$$

subsets.

```
library(MASS)
# importing data

boston_data <- read.csv("Boston.csv")

data_use <- dplyr::select(boston_data, high_temp_f,
high_dew_point_f, avg_humidity_percent, avg_visibility_mi,
avg_wind_mph, precip_in)

full_model <- lm(precip_in ~ ., data = data_use)
null_model <- lm(precip_in ~ 1, data = data_use)

pander(summary(stepAIC(null_model, direction = "forward",
scope = list(upper = full_model,
lower = null_model))))
```

Start: AIC=-8899.99 precip\_in ~ 1

|                        | Df | Sum of Sq | RSS    | AIC      |
|------------------------|----|-----------|--------|----------|
| • avg_visibility_mi    | 1  | 109.247   | 239.64 | -10306.2 |
| • avg_humidity_percent | 1  | 63.001    | 285.89 | -9644.6  |
| • high_dew_point_f     | 1  | 11.012    | 337.87 | -9018.2  |
| • avg_wind_mph         | 1  | 8.730     | 340.16 | -8993.0  |
|                        |    |           | 348.89 | -8900.0  |

- high\_temp\_f 1 0.009 348.88 -8898.1

Step: AIC=-10306.17 precip\_in ~ avg\_visibility\_mi

|  | Df | Sum of Sq | RSS | AIC |
|--|----|-----------|-----|-----|
|--|----|-----------|-----|-----|

- avg\_wind\_mph 1 4.7913 234.85 -10380
- high\_dew\_point\_f 1 4.6229 235.02 -10377
- avg\_humidity\_percent 1 3.3433 236.30 -10357
- high\_temp\_f 1 2.4624 237.18 -10343 239.64 -10306

Step: AIC=-10379.88 precip\_in ~ avg\_visibility\_mi + avg\_wind\_mph

|  | Df | Sum of Sq | RSS | AIC |
|--|----|-----------|-----|-----|
|--|----|-----------|-----|-----|

- high\_dew\_point\_f 1 7.7944 227.05 -10504
- avg\_humidity\_percent 1 6.2082 228.64 -10478
- high\_temp\_f 1 4.4022 230.44 -10449 234.85 -10380

Step: AIC=-10504.42 precip\_in ~ avg\_visibility\_mi + avg\_wind\_mph + high\_dew\_point\_f

|  | Df | Sum of Sq | RSS | AIC |
|--|----|-----------|-----|-----|
|--|----|-----------|-----|-----|

- high\_temp\_f 1 1.7389 225.31 -10531
- avg\_humidity\_percent 1 1.6070 225.45 -10529 227.05 -10504

Step: AIC=-10531.24 precip\_in ~ avg\_visibility\_mi + avg\_wind\_mph + high\_dew\_point\_f + high\_temp\_f

|  | Df | Sum of Sq | RSS | AIC |
|--|----|-----------|-----|-----|
|--|----|-----------|-----|-----|

- avg\_humidity\_percent 1 0.17494 225.14 -10532 225.31 -10531

Step: AIC=-10532.16 precip\_in ~ avg\_visibility\_mi + avg\_wind\_mph + high\_dew\_point\_f + high\_temp\_f + avg\_humidity\_percent

|                      | Estimate  | Std. Error | t value | Pr(> t )   |
|----------------------|-----------|------------|---------|------------|
| (Intercept)          | 0.4519    | 0.06116    | 7.39    | 1.802e-13  |
| avg_visibility_mi    | -0.07122  | 0.002693   | -26.44  | 2.062e-141 |
| avg_wind_mph         | 0.01305   | 0.001101   | 11.85   | 7.645e-32  |
| high_dew_point_f     | 0.004363  | 0.001085   | 4.021   | 5.897e-05  |
| high_temp_f          | -0.002076 | 0.000919   | -2.259  | 0.02396    |
| avg_humidity_percent | 0.001071  | 0.0006282  | 1.705   | 0.0882     |

Table 5: Fitting linear model: precip\_in ~ avg\_visibility\_mi + avg\_wind\_mph + high\_dew\_point\_f + high\_temp\_f + avg\_humidity\_percent

| Observations | Residual Std. Error | $R^2$  | Adjusted $R^2$ |
|--------------|---------------------|--------|----------------|
| 3749         | 0.2453              | 0.3547 | 0.3538         |

## 2.2.2 Backward Selection

The **backward selection** algorithm begins with the full set of variables.

Eliminate the least significant variable (the one with the largest p-value)

Stop if:

- All variables are significant, or
- All variables have been eliminated (intercept-only model).

```
library(MASS)
# importing data

boston_data <- read.csv("Boston.csv")

data_use <- dplyr::select(boston_data, high_temp_f,
high_dew_point_f, avg_humidity_percent, avg_visibility_mi,
avg_wind_mph, precip_in)

full_model <- lm(precip_in ~ ., data = data_use)
null_model <- lm(precip_in ~ 1, data = data_use)

pander(summary(stepAIC(null_model, direction = "backward",
scope = list(upper = full_model,
lower = null_model))))
```

Start: AIC=-8899.99 precip\_in ~ 1

|             | Estimate | Std. Error | t value | Pr(> t )   |
|-------------|----------|------------|---------|------------|
| (Intercept) | 0.1099   | 0.004983   | 22.06   | 1.701e-101 |

Table 7: Fitting linear model: precip\_in ~ 1

| Observations | Residual Std. Error | $R^2$ | Adjusted $R^2$ |
|--------------|---------------------|-------|----------------|
| 3749         | 0.3051              | 0     | 0              |

### 2.2.3 Stepwise Selection

At each iteration, the Step wise Selection (SW) algorithm consider all models obtained by either adding or deleting one term to or from the current model.

At each iteration, choose the model with the lowest AIC or BIC

Stop if the model with the lowest AIC/BIC is the current model

Otherwise, let the model with the lowest AIC/BIC be the new current model and then go back to Step 1

## 2.3 Model Selection

We use the BIC and AIC to select the optimal model.

The model with the smallest AIC and BIC is the best model.

AIC model:

$$AIC = 2k - 2 \ln(\hat{L})$$

- AIC: Akaike Information Criterion
- k: number of estimated parameters in the model
- $\hat{L}$ : maximum value of the likelihood function for the model

$$BIC = 2 \times L + d \log N$$

- N: Sample size
- d: total number of parameters